

Unbiased Estimator

Subjective Problems

Q.1.

Prove that the sample mean is an unbiased estimator of population mean:

$$E(\bar{x}) = \mu$$

Q2.

Prove that:

$$E\left(\frac{1}{n} \sum (x_i - \bar{x})^2\right) = \frac{n-1}{n} \sigma^2$$

Hence show that it is a biased estimator.

Q3. If T is an unbiased estimator for θ , show that T^2 is a biased estimator for θ^2

Q4.

Q. If $x_1, x_2, \dots, x_n$ be a random sample of size n from Normal population with mean μ and variance σ^2 then check the biasedness of $t = \frac{1}{n} \sum_{i=1}^n x_i^2$ over μ^2 .

Q5. If X is a binomial random variable, then discuss the biasedness and consistency of $T_1 = \frac{X}{n}$ and

$$T_2 = \frac{1}{(n+\sqrt{n})} \left(X + \frac{\sqrt{n}}{2}\right)$$

Q6. Let X be distributed in the Poisson form with parameter θ . Show that $T(X) = (-k)^X$ is an unbiased estimator of $e^{-(k+1)\theta}$, $k > 0$

MCQs Problems

Q1. A sample consists of 5, 7, 9, 11, 13. Find the estimator of population mean.

A. 7

B. 8

C. 9

D. 10

Answer: C. 9

Q2. For the sample 2, 4, 6, 8, find the sample variance.

A. 5.33

B. 6.67

C. 7.50

D. 8.00

Answer: B. 6.67

Q3. A sample mean is 50 from a sample of size 25. Which parameter is estimated?

A. Population variance

B. Population mean

C. Population median

D. Population mode

Answer: B. Population mean

Q4. If $E(\hat{\theta}) = \theta$, the estimator is called:

A. Biased

B. Efficient

C. Unbiased

D. Inconsistent

Answer: C. Unbiased

Q5. For $n=100$ and $s=20$, find the standard error.

A. 1

B. 2

C. 3

D. 4

Answer: B. 2

Q6. A sample mean of 18 estimates:

A. Population mean

B. Population variance

C. Population proportion

D. Population median

Answer: A. Population mean

Q7. A Poisson sample mean is 10. Find λ .

A. 8

B. 9

C. 10

D. 11

Answer: C. 10

Q8. A sample of size 16 has $s=12$. Find standard error.

- A. 2
- B. 3
- C. 4
- D. 5

Answer: B. 3

Q9. Sample values are 10, 20, 30. Find mean estimator.

- A. 15
- B. 18
- C. 20
- D. 25

Answer: C. 20

Q10. Sample 1, 3, 5, 7, 9. Find sample mean.

- A. 4
- B. 5
- C. 6
- D. 7

Answer: B. 5

Q11. If estimator variance decreases as sample size increases, the property is:

- A. Efficiency
- B. Consistency
- C. Sufficiency
- D. Bias

Answer: B. Consistency

Q12. For sample 8, 10, 12, 14, find mean.

- A. 9
- B. 10
- C. 11
- D. 12

Answer: C. 11